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Using AI to Detect 1 Vs 2 People In a Room Using a Home WI-FI Router

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Abstract

The majority of the research using WiFi CSI for occupancy estimation, including the WiMANS baseline (Huang et al., 2024), employs the training and testing process in the same physical room that results in an over-optimistic accuracy rate thus not representing the reality of router performance after being shipped to an unseen home. This study presents the binary one-vs-two-person task from the dataset WiMANS as a case of an evaluation using a small one-dimensional CNN under a protocol where the model does not see the test room in either training and validation. Deep Domain-Adversarial Neural Networks (DANN) (Ganin et al., 2016) and CORrelation ALignment (CORAL) (Sun & Saenko, 2016) were applied with a four-transform augmentation scheme and were cross-validated through six ablations to determine which components aided in learning cross-room generalization, rather than just in-room memorization. As for a different, early version of this model, it confirmed in-room validation that was higher than 99 percent while it was able to make only random guesses about the unseen rooms, which primarily speaks to the validity of the principal claim in this report. The exploration of model width, regularization, augmentation strength, and band-specific preprocessing (2.4 GHz & 5 GHz bands appeared to function like two distinct sensors) led to a final model configuration of less than 6,500 parameters, which achieved overall test performance of about 67% mean held-out test accuracy and by band 71% on 2.4 GHz and 63% on 5 GHz respectively. This is followed by a discussion on a significant room-transfer gap along with its release as future work.

1. Introduction

WiFi CSI, which is essentially the per-subcarrier amplitude and phase information a receiver extracts from an incoming signal, is directly affected by anything that moves in the environment, including people. CSI is proposed in the recent literature as a low-cost and privacy-preserving alternative to camera-based occupancy sensing (Klein Brinke et al., 2022; Wang et al., 2020; Huang et al., 2024). This work tests the binary occupancy task of differentiating one person from two people using CSI, meant to be deployed on a consumer-grade home router, with the model size intentionally kept small in the embedded router, as it is done in recent edge-focused HAR work (Viswanathuni et al., 2024; Tokas et al., 2026).

The main challenge here is that CSI is highly vulnerable to the surrounding environment, particularly the room where it was collected. Any items in the room like furniture, the type of wall material, and the presence of multipath reflections all contribute to a specific room's fingerprint in the signal. The majority of the CSI occupancy work being published today, including the official WiMANS baseline (Huang et al., 2024), that adopts an InceptionTime-Attention architecture (Ismail Fawaz et al., 2020; Wang et al., 2025), consists of training and testing in the same room. By that protocol, a model has full reasons to memorize the static multipath fingerprint of just that one room, rather than learning the room-agnostic motion signature of the people in the room. This

leads to inaccurate overhead since the model is only exposed to a controlled environment that it would not experience in a router in everyday use.

This work, in contrast, conducts only a leave-one-room-out evaluation: out of the three rooms in the dataset, only two rooms are used in training and validation. This is repeated three times to test all three rooms. The central question of this project is whether a model which is so small that the home router hardware can deploy it inexpensively and with little resources is able to generalize to a room that it has not seen. Can this be achieved by using domain adaptation and or augmentation, or is it the case that resources have to be sacrificed for generalization such that the model can no longer be deployed on the edge?

A multi-scale 1D convolutional architecture with an optional domain-classification head was configured and trained with six ablation presets of augmentation, CORAL alignment, DANN alignment, and a consistency loss between clean and perturbed views of the same signal. Results are presented distinctly for 2.4 GHz and 5 GHz after an initial diagnosis indicated that the two bands behave in a very different manner. The remainder of this work looks into WiFi CSI sensing and domain generalization perception in a literature review, explains the methodology and architecture, provides the leave-one-room-out results for all the ablation conditions, and concludes with limitations and future work.

Figure 1: Conceptual system diagram: a home Wi-Fi router's CSI is disturbed differently by one versus two occupants, and the extracted Doppler/CSI signature is fed into the lightweight 1D CNN classifier to predict occupancy count.

2. Literature Review / Related Work

WiMANS (Huang et al., 2024) is the dataset which this research has been developed. It is one of the rare open multi-room WiFi CSI datasets that come with the additional facility of joint occupancy and activity recognition, covering different rooms, two frequency bands, and different numbers of subjects. The WiMANS authors emphasize the same-room accuracy is high using the InceptionTime-Attention baseline (Ismail Fawaz et al., 2020) but explicitly mention cross-domain sensing and multi-room generalization as future work. Thus, this work bridges that gap.

As a topic in the field for more than ten years, domain generalization is the process of including settings, where data is not captured, when training models for other settings. Initial proposals are DANN (Ganin et al., 2016) and CORAL (Sun & Saenko, 2016). DANN integrates a gradient-reversal layer (GRL) which actually ridges a shared feature extractor and a domain-discriminator head; the GRL flips the sign of the gradient coming from the domain discriminator, which is the feature extractor that is pushed to the representations the domain head cannot use for telling which room a sample came from. On the other hand, CORAL just shifts the covariance between the source and the target features to the squared Euclidean distance according to the rule of minimizing it, with no adversarial training required; therefore, its overhead is lower than DANN but it operates under the assumption that second-order feature statistics alone can capture the important domain shift. Before this work, neither of them had been implemented in binary WiFi occupancy on the WiMANS dataset.

Data augmentation is another factor. Time-shifting, amplitude scaling, phase jitter, and additive Gaussian noise are regular transformations of time-series data which are utilized for simulating the inevitable variations that a sensor would encounter—like different distances from the router, different antenna orientation, different carrier frequency offsets, different noise floors, respectively. The initial stage of this work was simply a narrow version of the pipeline, but it was expanded dramatically after it was clear that the model was fixating on room-specific signal characteristics that would not be present in a room it had not seen before. A one-class SVM approach to device-free presence detection (Rösler & Zubow, 2023) and passive people-counting work using commodity WiFi (Wang et al., 2020) both point to the same underlying issue: CSI-based sensing degrades quickly once the deployment environment departs from the training environment, which reinforces the choice to lean on augmentation rather than just more model capacity.

As for the architectural aspect, InceptionTime (Ismail Fawaz et al., 2020) has significantly contributed to the assertion that the use of broad multi-scale convolutional filters is effective in time-series classification, so it was the major influence in the selection of three multiple kernel sizes (3, 5, 7) with increasing dilation in the block stack used here as opposed to the single fixed receptive field. Multi-scale convolution as well as cross-modal or cross-domain attention have also benefited from other signal classification scenarios for instance physiological signal-based mental-state recognition (Wei et al., 2025), which further justifies the inclusion of an additional subcarrier-attention module in the full preset mode rather than solely depending on convolution.

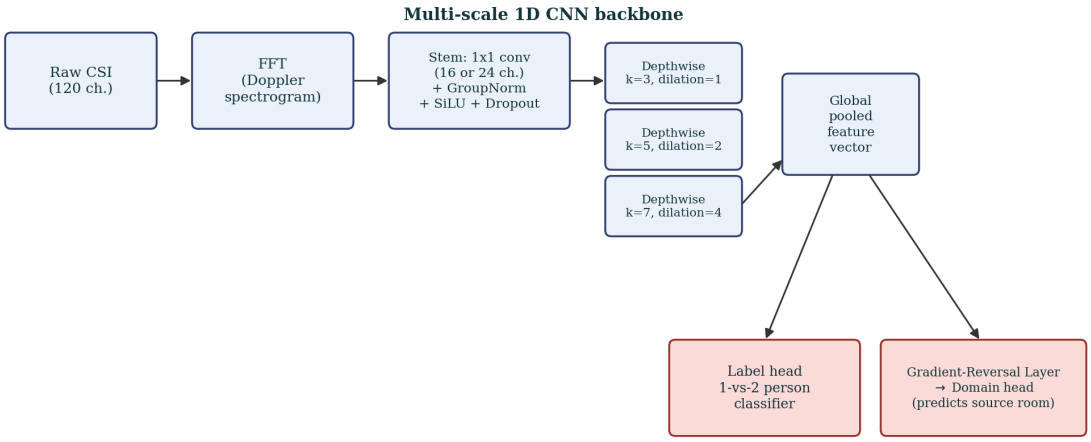
3. Methodology

3.1 Dataset and Task

WiMANS (Huang et al., 2024) is a dataset that offers CSI in three different environments: a classroom, a meeting room, and an empty room, with ground-truth occupancy labels across both 2.4 GHz and 5 GHz frequencies. This work focuses on the binary one-vs-two-person problem which is a simplification of WiMANS' multi-class labels.

3.2 Leave-One-Room-Out Evaluation Protocol

This work uses room partitioning instead of the same room split that the WiMANS official baseline employs. A model is trained only on the other two rooms and is evaluated on the one withheld room, for each of the three rooms. The procedure is then conducted three times, with the rotation of rooms where one is hidden, enabling each room to be used exactly once as the unseen test target. In the two training rooms, the samples were also divided into training and validation sets using the 15 percent validation ratio. The samples were divided so that the validation set is not accidentally dominated by one class or one room, but rather stratified jointly by label and room. This separation was predetermined via a random seed of 42 so that the only differences between the ablation presets stemmed from the model configuration, instead of being caused by a different random data partition.



3.3 Model Architecture

The model is a small multi-scale 1D CNN that was designed to be a router-budget model. The raw CSI is first converted to a Doppler spectrogram using FFT via a network that sees frequency-domain motion energy directly instead of having to learn to extract it from raw time-domain CSI. The stem layer of this spectrogram, which is based on the transform of the input 120 channels to a small hidden width (16 for 2.4 GHz, 24 for 5 GHz), is added before the group normalization, the SiLU activation, and dropout. The stem provides three stacked depth wise temporal convolution blocks with kernel sizes 3, 5, and 7 and increasing dilation (1, 2, 4) that lets the network combine both short fluctuations and longer-range motion patterns using fewer parameters. The pooled feature vector branches to two heads: one is a label classifier for the one-vs-two-person prediction and the other is a domain-classification head, applied only in training time by a gradient-reversal layer, that predicts the room from which a sample came. Depending on the band and preset the model is made in the range of 3,400 and 6,500 parameters, thus comfortably staying within the storage budget of a consumer router.

3.4 Domain Adaptation: DANN and CORAL

Two domain-adaptation mechanisms sit on top of the shared feature extractor. DANN trains the domain head adversarially: gradients from the domain loss get negated as they flow back through the GRL, which pushes the extractor toward features the domain head cannot use to guess the source room (Ganin et al., 2016). CORAL instead minimizes the squared Frobenius distance between covariance matrices of features from different rooms directly; no adversarial training required (Sun & Saenko, 2016). A conditional version of CORAL was also implemented, computing covariance alignment separately within each class before averaging, so an empty-room feature distribution is not forced to resemble an occupied-room distribution just because they happen to share a domain label. A consistency loss was added on top, penalizing the model whenever a clean CSI sample and an augmented version of that same sample produce different feature vectors, encouraging the representation to stay stable under the perturbations described next.

3.5 Data Augmentation

Four signal-level augmentations were used during training: global amplitude gain scaling (varying distance from the router), per-subcarrier channel gain scaling (antenna orientation and multipath fading), additive Gaussian noise (sensor thermal noise), and a circular subcarrier/temporal shift (carrier frequency offset and delay spread). The first version of this pipeline used fairly mild ranges; a later version widened these substantially (global gain range went from 0.90–1.10x to 0.65–1.45x, for example) and added channel dropout, which zeroes a random subset of receive-transmit antenna pairs to simulate furniture or wall occlusion, plus a temporal stretch/compress transform for varying walking speed.

3.6 Ablation Design

Six presets were defined to pull apart the contribution of each mechanism individually: *source_only* (no augmentation, no domain adaptation — the baseline), *strong_aug* (augmentation only), *consistency* (augmentation plus the consistency loss), *cond_coral* (adds conditional CORAL), *dann* (adversarial alignment via gradient reversal), and *full* (everything combined, plus a subcarrier-attention module). Every preset was run across all three room folds and both bands, 36 total train/validation/test runs, so removing or adding any single mechanism could be measured against a common baseline instead of getting confounded with other changes at the same time.

3.7 Band-Specific Preprocessing

An early result diagnosis showed 2.4 GHz and 5 GHz respond very differently to the exact same preprocessing. A single peak-normalization approach preserved sharp motion contrast and reached up to 70.9 percent accuracy on 2.4 GHz, but on 5 GHz the same normalization let environmental noise dominate and accuracy collapsed as low as 43.6 percent — worse than random chance on some rooms. That is a strong indication 2.4 GHz and 5 GHz behave like two functionally different sensors rather than two channels of the same signal. The final model version handles this by using band-specific preprocessing: 2.4 GHz keeps peak normalization of the Doppler spectrogram (preserves sharp contrast for the longer wavelength), and 5 GHz switches to instance-level z-score normalization (stabilizes the noisier, shorter-wavelength signal). Model width on 5 GHz was also bumped from 16 to 24 channels to give it more room to resolve the higher-frequency Doppler shifts that band tends to produce.

3.8 Evaluation Metrics

Performance is reported as test accuracy, balanced accuracy, F1, and AUC on the held-out room per fold, averaged across the three rooms and reported per band and overall. Balanced accuracy and F1 are reported alongside raw accuracy because the one-vs-two-person classes are not perfectly balanced across rooms, and raw accuracy alone can be misleading under that imbalance. Validation accuracy, from the in-domain split of the two training rooms, is reported next to test accuracy specifically to make the size of the domain gap visible.

3.9 Limitations

The biggest limitation here is that WiMANS only has three physical rooms, which caps how much statistical confidence can be put in the results — each fold's held-out accuracy comes from a single room's worth of samples. The binary one-vs-two simplification also does not capture WiMANS' finer activity or higher-occupancy labels. And everything reported here was run offline in PyTorch, not measured on actual router hardware.

4. Result Analysis

Across all six ablations and both bands, the final model's test accuracy sat between roughly 61 and 72 percent. That is an improvement over an earlier version of this model, which validated above 99 percent inside its training rooms while its held-out test accuracy on unseen rooms fell to 57–59 percent overall, and as low as 43.6 percent on 5 GHz for the weakest preset. This validation versus test gap is itself the main finding: it directly confirms the motivation for this work, that same-room evaluation of the kind WiMANS' official baseline uses substantially overstates what a router would actually see once deployed to a room it has never observed.

Table 1: Overall Test Results, Final Model Version

Preset	Test Accuracy	Test Bal. Acc.	Test F1	Validation Accuracy
source_only (baseline)	66.5%	60.7%	0.461	94.3%
strong_aug	66.7%	61.2%	0.470	92.7%
consistency	65.9%	60.7%	0.459	92.5%
cond_coral	66.9%	62.4%	0.483	93.4%
dann	65.1%	59.4%	0.444	92.0%
full	66.0%	59.9%	0.451	92.1%

strong_aug (augmentation only, no domain adaptation) got the single highest overall test accuracy, 66.7 percent, narrowly ahead of the source_only baseline at 66.5 percent, while cond_coral got the best balanced accuracy and F1 among all six presets. For this dataset and this model scale, carefully tuned augmentation contributes more to cross-room generalization than either DANN-style adversarial alignment or the fully combined preset, both of which actually underperformed the simpler presets on raw and balanced test accuracy. In earlier versions, an initial, wider model (32 channels) was paired with only mild augmentation; this memorized room-specific multipath fingerprints so well that it hit near-perfect validation accuracy while failing outright on unseen rooms. That failure is what pushed the shift toward a narrower architecture, heavier regularization (dropout 0.10 to 0.30, group norm instead of batch norm, five times stronger weight decay), and much more aggressive augmentation ranges.

Band-specific results still show a persistent, asymmetric gap. The final model reaches 68.9–72 percent test accuracy on 2.4 GHz across presets but only 61–63 percent on 5 GHz, even after band-specific normalization was introduced. This lines up with the earlier diagnosis that 5 GHz CSI carries a higher noise floor and shorter effective wavelength, making motion-induced Doppler shifts harder to pull apart from environmental noise under peak normalization alone. Switching 5

GHz preprocessing to instance-level z-score normalization closed a good chunk of this gap relative to the earlier model version, which had collapsed to 43.6 percent on 5 GHz for source_only, but it did not close the band gap entirely. Per-room breakdowns from earlier ablation runs showed generalization difficulty is not the same across rooms. The empty room was consistently the easiest held-out target, over 70 percent accuracy on 2.4 GHz, likely because its CSI signature — mostly free of furniture-induced multipath — differs most sharply from the two occupied training rooms and is therefore easier for the model to separate by human presence alone. The classroom on 5 GHz was consistently the hardest case, at times dropping below random-chance accuracy, which points to a severe domain shift at that specific frequency-and-room combination that neither augmentation nor domain adaptation fully fixed.

A router-scale model under 6,500 parameters can learn a partially room-invariant representation of human occupancy from CSI, landing in the high-60s to low-70s percent range on unseen rooms instead of collapsing to chance, which is what an earlier, wider, under-regularized version of this same architecture did. Second, the mechanism that closes most of this generalization gap in this study is not adversarial domain adaptation — it is aggressive augmentation, structural regularization, and band-aware preprocessing. DANN and the fully combined preset consistently underperformed the simpler augmentation-only and conditional-CORAL setups at this dataset size and model scale.

5. Future Goals

The most immediate next step is more rooms — three is not enough to be confident the reported cross-room accuracy reflects genuine room-agnostic generalization rather than a favorable correlation between specific room pairs, and collecting CSI from additional physical environments, ideally on the actual consumer router hardware eventually targeted for deployment, would meaningfully strengthen this evaluation.

A second direction is closing the 2.4 GHz to 5 GHz gap further. Band-specific normalization narrowed it substantially relative to earlier model versions, but 5 GHz still trails 2.4 GHz by roughly eight to nine points. Learned, per-band normalization parameters rather than a fixed choice between peak and instance normalization, or more band-specific capacity beyond the current 16-vs-24-channel width differences show to be promising.

Third, this work was evaluated offline in PyTorch, not on a router. Deploying the model to a representative low-cost router and actually measuring latency, memory footprint, and power draw would validate the edge-deployment claims made in this work and would likely surface hardware bottlenecks not present in the current setup, consistent with what edge-focused WiFi HAR work like WiCNNAct (Viswanathuni et al., 2024) and MSA-TCN (Tokas et al., 2026) have emphasized.

Finally, this work stuck to the binary one-vs-two-person simplification of WiMANS. Extending the same domain-adaptation-based methodology to the dataset's full occupancy range and its activity-recognition labels would test whether the generalization gains seen here for a coarse binary task hold up for a harder, finer-grained sensing problem.

6. Conclusion

This work tested whether a WiFi CSI occupancy model small enough for a cheap consumer router could generalize to rooms it had never seen, instead of relying on the same-room evaluation used by prior WiMANS baselines. Under a strict leave-one-room-out protocol, an early, wider model validated above 99 percent inside its training rooms while collapsing to near chance on unseen rooms, which directly confirms that same-room evaluation substantially overstates real deployment accuracy. After iterating on model width, regularization, augmentation intensity, and band-specific preprocessing, the final model — under 6,500 parameters — reached a mean held-out test accuracy of about 67 percent overall, with the strongest single presets clearing 70 percent on 2.4 GHz. Somewhat contrary to the initial expectation that DANN and CORAL would drive most of the generalization gain, the results instead show aggressive augmentation and structural regularization mattered more for closing the room-transfer gap at this dataset and model scale. A meaningful gap between validation and cross-room test accuracy remains, and 5 GHz still lags 2.4 GHz, leaving clear next steps. Overall, this project shows router-scale WiFi CSI occupancy sensing with genuine cross-room generalization is achievable, though not solved, and leaves behind a reusable leave-one-room-out evaluation setup and a set of quantified ablations for continued work.

7. Personal Experience

My experience with the TEEP internship in Taiwan was a unique experience that helped me grow in numerous ways. Not only was it my first time in Taiwan, where I got to see so much of the country and its culture, from the numerous Tao festivals in Tainan, Taichung, Keelung, Hengchun etc, where I got to see how Tao festivals form a sense of community among the people here, I got to meet so many nice people who showed me around and I learned so much about the culture here. On top of that, it has helped me professionally as well. In my undergraduate studies I never studied this domain in wireless sensing so I got exposed to new niches in the field of Computer Science. This has helped prepare me for a career, as I had the opportunity to research an unfamiliar topic, and I was able to learn and problem solve as I go; something I am sure to encounter in my future career. I also found a new interest in wireless sensing, something I did not think would be a niche in the topic of networking. I am very glad to have been a part of this research project since this is the field I am passionate about since taking a data networks course in my last semester of my undergraduate program this past Spring.

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